

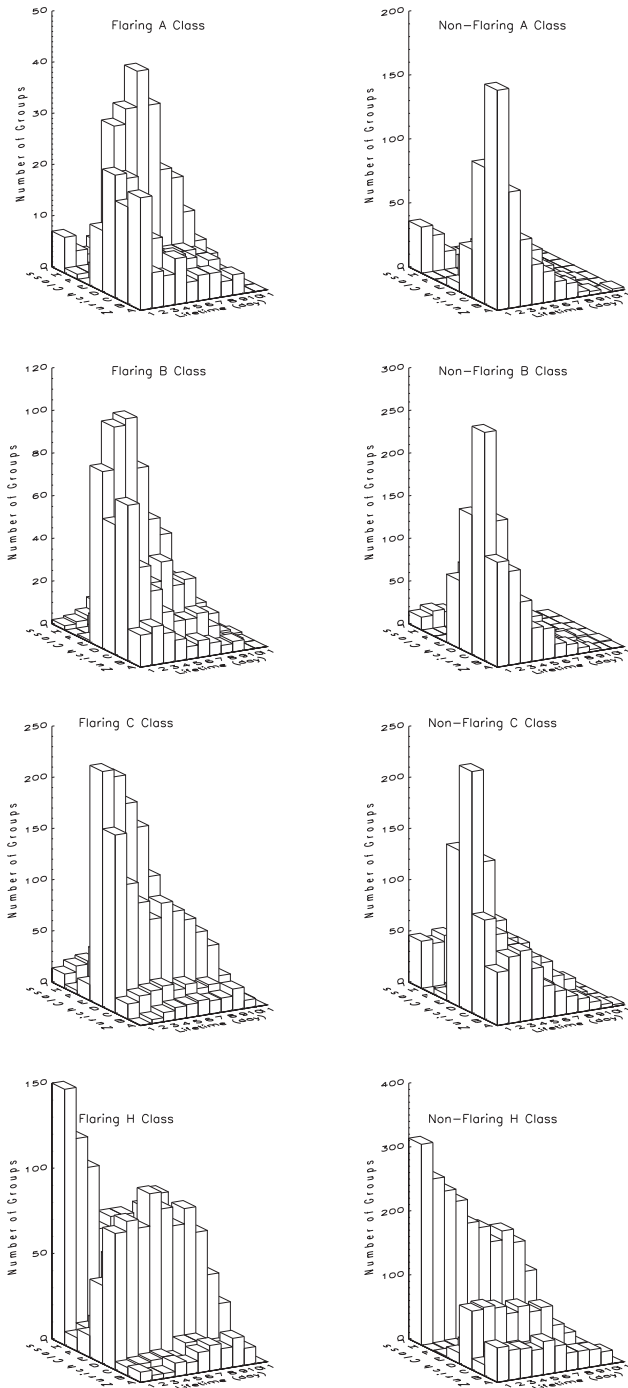


Figure 1. Evolution of Zurich class during the lifetime of ARs, plotted separately for flaring (left) and non-flaring (right) ARs for classes A, B, C and H of small ARs. See details in the text.

of non-flaring ARs (Figs 1 and 2, right), they mainly remain in the same classification during their lifetime. Note that their lifetimes are shorter than those of flaring ARs.

3.3 Flaring effect on sunspot groups

To investigate how flaring affects the class evolution of ARs, we considered all flaring episodes (each episode may include one or

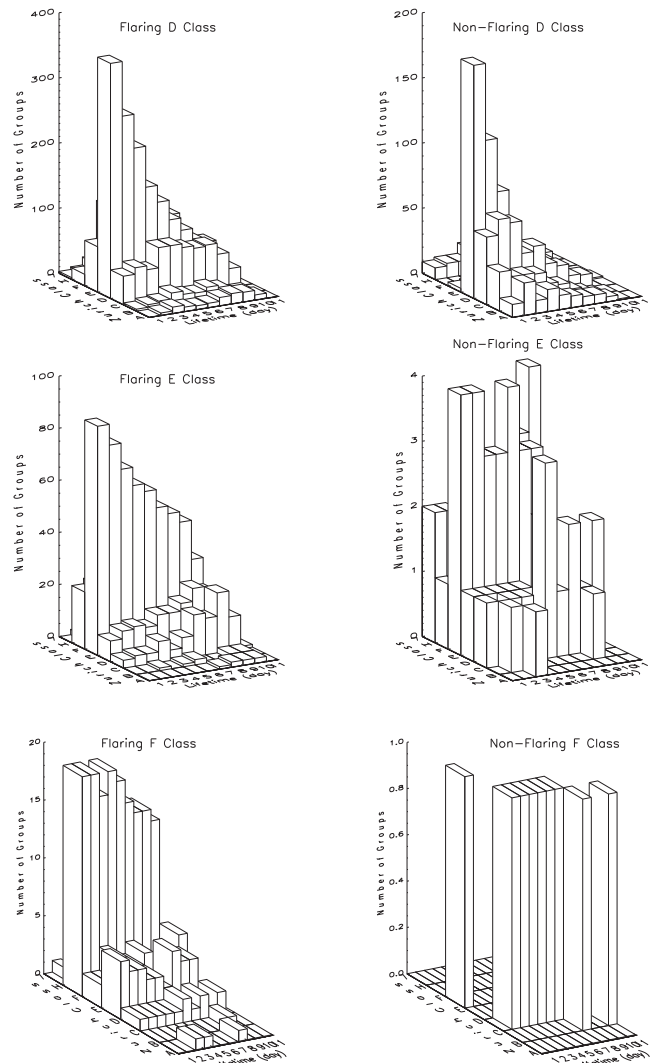


Figure 2. Evolution of Zurich class during the lifetime of ARs, plotted separately for flaring (left) and non-flaring (right) ARs for classes D, E and F of large ARs. See details in the text.

more flares) in all flaring ARs and followed the evolution of their class. Altogether, 4312 flaring episodes occurred in all classes of sunspot groups (Fig. 3).

For flaring ARs, the maximum in the distribution of flares occurs for class D. In McIntosh (1990), the maximum of the distribution is for groups of class F. This difference is probably because McIntosh (1990) took into account only strong flares of classes M and X. In our figure, all flares of class C₀ and higher were taken into account. On the other hand, this may be a specific trait of low activity in cycles 23 and 24.

We recorded changes in the modified Zurich class, average sunspot counts (SSCs) and sunspot areas (SSAs) that occurred between the flaring day and the day after the flare event (Table 2). The table shows the percentage of ARs of a given initial class (leftmost column) that changed class. For example, there were 53 flaring events in class A groups and there was no classification change in 36 percent of these ARs, while 64 percent of them changed (18.9 percent evolved to B, 22.6 percent to C, 18.9 percent to D and 3.8 percent to H). The general tendency seen in this table is that most of the class A and B groups rapidly evolved to higher classes (C and D). Starting from class C, the tendency changes and the